

Network Devices

Q1. Simple home network diagram

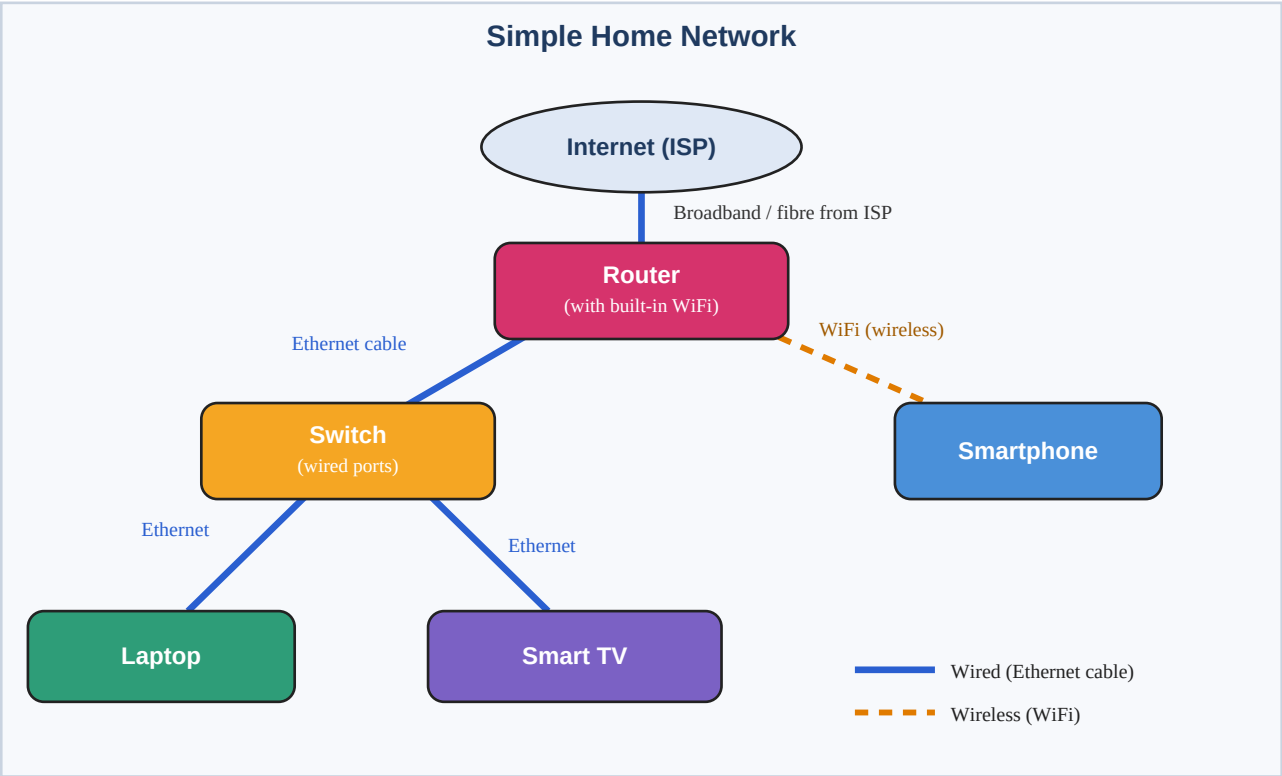


Figure 1: Home network with a router, a switch and three devices

Device	How it connects
Router	Connects to the ISP's line and joins my home network to the Internet. It also gives out WiFi and IP addresses.
Switch	Connected to the router by an Ethernet cable. It provides extra wired ports for devices.
Laptop	Wired: Ethernet cable to the switch.
Smart TV	Wired: Ethernet cable to the switch, for smooth streaming.
Smartphone	Wireless: WiFi to the router's built-in access point.

Q2. Comparison of network devices

Device	What it does	OSI layer	Real-world use case
Router	Connects different networks and forwards packets between them using IP addresses. It links my home to the Internet.	Layer 3 (Network)	The router at home gives my phone Internet access to watch YouTube or order on Zomato.
Switch	Connects wired devices in one LAN and sends data only to the port of the intended device, using MAC addresses.	Layer 2 (Data Link)	Connecting many PCs in a college computer lab or a LAN gaming session so all can share files or play together.
Hub	Repeats incoming data to every port, so all devices receive it. It is simple but causes collisions and slow speed.	Layer 1 (Physical)	Old small LANs, such as an early cyber cafe network for multiplayer games. Now replaced by switches.
Access point	Adds WiFi to a wired network so wireless devices can join it.	Layer 2 (Data Link)	College or cafe WiFi where I use Instagram or WhatsApp on my phone without any cable.

Q3. How data travels from my phone when I order food on Zomato

I am at home and tap **Place Order** in the Zomato app. My phone packs the order into data and sends it as WiFi radio signals. The **access point** (built into my router) receives the signals and turns them into wired network data. If my home has a separate **switch**, it forwards this data to the router, because it knows which port the router is on. The **router** then sends the data to my ISP, which passes it across the Internet to Zomato's server. The server confirms the order and sends the reply back along the same path, and my phone shows "Order placed".

Device	Role in this order
Access point	Receives the phone's WiFi signal and connects it to the wired network.
Switch	Passes the data from the access point to the router using MAC addresses, without sending it to other devices.
Router	Sends the data out to the ISP and the Internet, and directs Zomato's reply back to my phone.

In most homes the router, switch and access point are combined in one box.

Q4. Connecting a far-away gaming PC that has only an Ethernet port

The PC cannot use WiFi by itself, so I need a device that carries the router's signal to a spot near the PC. Two good options are:

Option	How it works in this setup	Points to note
1. Powerline adapter kit	One adapter plugs into a wall socket near the router and connects to it by Ethernet cable. The second adapter plugs into a socket near the PC and connects to the PC by Ethernet. The network data travels through the home's electrical wiring between the two adapters.	Steadier than WiFi for gaming. Speed depends on the quality of the wiring, and both sockets should be on the same electrical circuit.
2. WiFi extender with an Ethernet port	The extender is placed midway where it still gets a good WiFi signal from the router. It repeats that WiFi, and the PC is plugged into the extender's Ethernet port with a short cable, so the PC gets the Internet through the extender.	Easy and needs no new cable. Speed is usually reduced and latency is slightly higher, so it is less ideal for competitive gaming.

Best option if possible: run a long Cat6 Ethernet cable from the router (or a switch near it) to the PC, which gives the fastest and most reliable connection.